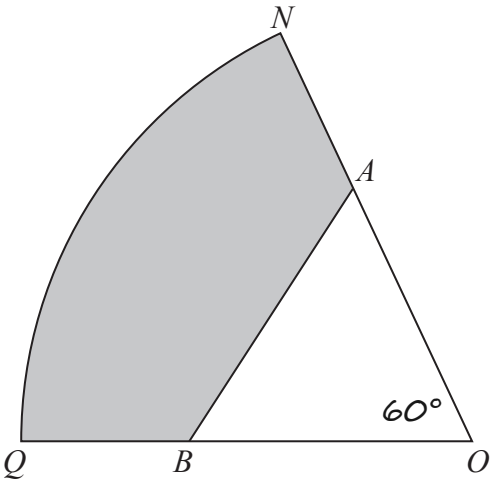
17



ONQ is a sector of a circle with centre *O* and radius 11 cm.

A is the point on *ON* and *B* is the point on *OQ* such that *AOB* is an equilateral triangle of side 7 cm.

Calculate the area of the shaded region as a percentage of the area of the sector *ONQ*.
Give your answer correct to 1 decimal place.

Equilateral triangle: angle AOB = 60°.

1 mark
Use 60°

$$\text{Sector area } S = \frac{60}{360} \times \pi \times 11^2 = 63.35545\ldots \text{ cm}^2$$

1 mark
Use the sector fraction of the full circle

$$\text{Triangle area } T = \frac{1}{2} \times 7 \times 7 \times \sin 60° = 21.21762\ldots \text{ cm}^2$$

1 mark
Use ½ab sin C for the triangle

$$\text{Shaded area} = S - T = 42.13783\ldots \text{ cm}^2$$

$$\text{Percentage} = \frac{S - T}{S} \times 100 = 66.51019\ldots\%$$

1 mark
Shaded area ÷ whole sector area × 100

Keep unrounded values: 66.5% to 1 decimal place.

1 mark
Correct percentage, rounded at the end

66.5 ..... %

(Total for Question 17 is 5 marks)